Course milestone M17: Reproducible Package and the Peer Cold Run

HONR 46400 · Evidence-Driven Research

Start here

Your main file is `lastname_m17.pdf`; the artifact listed at the end of this section travels beside it. Build it in this order:

1. **Your carry-forward record, at the very top.** Open my review email and your research action plan side by side. Answer every numbered request for this milestone (each has an ID like `IR-M17-01`) and every step your plan sets for this milestone. Next to each one, write **Done** and exactly where I can find it, or **Not adopted**, with your reason and what you chose instead. Several items can point to the same piece of work.
2. **Then your milestone work and files.** The book milestone below explains the research work. The complete course checklist, the files to hand in and the rubric are in this milestone's assignment on Brightspace: check them before you submit.

How this counts. Your answer to the action plan is worth 10 points of the rubric, scored for you individually. Each numbered request left unanswered costs 5 points, up to 20. Only requests and plan steps I sent you at least 48 hours before the deadline count; anything later moves to the next milestone. A step that is also a numbered request is counted once, as a request. If I sent you no numbered requests, or no plan steps apply, say so in one line. Not sure what applies to you? Write to me before the deadline; that is exactly what I am here for.

Submit it on Brightspace, under **Assignments** then **M17**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 11: Your reproducible package** (checkpoint, version 1) of the course book, EDR| AI.

Beside your PDF, hand in:

- `lastname_m17_package.zip`: Your repaired reusable package (or a shared folder or repository link). It must run top to bottom in a clean session, so it travels as its own artifact.

Open the milestone workbook in Colab

This chapter closes Studio 11: Reproduce and package. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. An independent cold-run record and a reusable research package. Working alone, the record is a clean-environment rerun labeled "solo proxy; external cold run pending": the label, and the honesty it enforces, are part of the artifact.

What you bring

You also carry forward the last milestone's artifact: **Milestone 10: Your artifact, ready to publish or present**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 36: Replication and Reproduction: your cold reproduction record: every number re-run from the top, one undisclosed choice varied, and missingness classified before any filling.
- Lesson 37: Open and Reusable Research Packages: your research capsule: the runnable notebook, the data-provenance note with its terms of use, the fixed seed, the decision log, your AI-use ledger, the environment record, the README, and a cold rerun in a second environment.

The practice

1. Assemble the draft package first: data or its access route, code, environment, documentation, and licence.
2. Run your own package cold, from a clean environment, following only your written instructions.
3. Then the external test: have someone else run it without your help, and log every place they had to ask you a question. Working alone? Rerun it yourself after a real break, instructions only, and label the record 'solo proxy; external cold run pending': a proxy never becomes a claim of independent reproduction.
4. Fix what the runs exposed: every question asked is a missing line in your documentation, and freeze the final manifest with its date.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** What can be shared is set by your permissions, and the package says so explicitly.
- **Evidence, provenance, and reproducibility.** Reproduction is the evidence rail's final test.
- **AI activity, verification, and human decisions.** An assistant can generate documentation; only a cold human run can validate it.
- **Uncertainty, claim boundary, and revision history.** Reproduce the uncertainty statement, not only the point estimate.

Where this milestone sits

You arrived carrying Milestone 10: Your artifact, ready to publish or present. Milestone 12 decides whether the packaged work is ready to leave your hands, and what the next cycle should ask. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

A failed cold run is a finding about your work, and it precedes any release.

The book's own checks

Your course rubric on Brightspace scores these checks inside its rows, so use this table to test your work before you hand it in. The points below are the book's, not your course score.

Each row scores **0**, **1**, or **2**. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning

#	Criterion	0	1	2
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: An independent cold-run record (working alone: a clean-environment rerun labeled 'solo proxy; external cold run pending') and a reusable research package	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** `Runtime` ⟶ `Run all`, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

`File` ⟶ `Print` opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.